

Rigorous Formulation and Proof Strategy for the Erdős-Straus Conjecture

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Abstract

This document presents a rigorous axiomatic foundation, literature context, modular identity strategies, and an explicit mathematical derivation concerning the Erdős-Straus conjecture. Additionally, a structural architecture for autoformalization in Lean 4 is provided.

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1 Axiomatic Definitions

Definition 1.1. Let $\mathbb{N}_{>1} = \{n \in \mathbb{N} \mid n \geq 2\}$. The Erdős-Straus conjecture postulates that for all $n \in \mathbb{N}_{>1}$, there exist $x, y, z \in \mathbb{N}_{>0}$ such that:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \quad (1)$$

Definition 1.2. By multiplying the aforementioned equation by $nxyz$, we obtain the equivalent polynomial Diophantine formulation:

$$4xyz = n(xy + xz + yz) \quad (2)$$

2 Contextual Literature and Related Works

The Erdős-Straus conjecture, formulated in 1948 by Paul Erdős and Ernst G. Straus, seeks an expansion of $4/n$ into exactly three positive Egyptian fractions. Relevant literature extensively explores modular identities. For instance, Mordell (1967) demonstrated that an identity providing solutions for all $n \equiv r \pmod{p}$ exists if and only if r is not a quadratic residue modulo p . Computer searches (e.g., by Salez, 2014) have verified the conjecture up to $N = 10^{17}$. Elsholtz and Tao (2013) bounded the average number of solutions polylogarithmically using the Bombieri-Vinogradov and Brun-Titchmarsh theorems.

The Hasse principle suggests that studying Diophantine equations modulo p might piece together integer solutions. However, a complete covering system of congruences is proven impossible due to 1 being a quadratic residue modulo any $n > 1$.

3 Strategy of Proof and Lemmas

The problem can be reduced to considering only prime values of n . If $n = m \cdot p$ where p is prime, a solution for p yields a solution for n by scaling the denominators by m .

Lemma 3.1 (Modular Case $n \equiv 2 \pmod{3}$). *For any integer $n > 1$ such that $n \equiv 2 \pmod{3}$, there exists a solution $(x, y, z) \in \mathbb{N}_{>0}^3$ satisfying $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.*

Lemma 3.2 (Modular Case $n \equiv 3 \pmod{4}$). *For any integer $n > 1$ such that $n \equiv 3 \pmod{4}$, there exists a solution $(x, y, z) \in \mathbb{N}_{>0}^3$ satisfying $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.*

4 Informal Mathematical Proof (Zero Ellipse)

We proceed to prove the first lemma explicitly.

Proof of Lemma 3.1. Let $n \in \mathbb{N}_{>1}$ be such that $n \equiv 2 \pmod{3}$. By definition of congruence modulo 3, there exists an integer $k \geq 0$ such that $n = 3k + 2$. We define $x = n$. Since $n \geq 2$, $x \in \mathbb{N}_{>0}$. We define $y = \frac{n+1}{3}$. We must verify that y is an integer. Substitute $n = 3k + 2$ into the expression for y :

$$\begin{aligned} y &= \frac{(3k + 2) + 1}{3} \\ &= \frac{3k + 3}{3} \\ &= k + 1 \end{aligned}$$

Since $k \geq 0$, $k + 1$ is an integer and $y \in \mathbb{N}_{>0}$. We define $z = \frac{n(n+1)}{3}$. We must verify that z is an integer. Substitute $n = 3k + 2$:

$$\begin{aligned} z &= \frac{(3k+2)(3k+3)}{3} \\ &= (3k+2)(k+1) \end{aligned}$$

Since $k \geq 0$, $3k + 2 \geq 2$ and $k + 1 \geq 1$. Both are integers, so their product is an integer, and $z \in \mathbb{N}_{>0}$.

We now compute the sum $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$:

$$\begin{aligned} \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{n} + \frac{1}{\frac{n+1}{3}} + \frac{1}{\frac{n(n+1)}{3}} \\ &= \frac{1}{n} + \frac{3}{n+1} + \frac{3}{n(n+1)} \end{aligned}$$

To add these fractions, we find a common denominator, which is $n(n+1)$.

$$\begin{aligned} \frac{1}{n} &= \frac{n+1}{n(n+1)} \\ \frac{3}{n+1} &= \frac{3n}{n(n+1)} \\ \frac{3}{n(n+1)} &= \frac{3}{n(n+1)} \end{aligned}$$

Summing the numerators:

$$\begin{aligned} \frac{n+1+3n+3}{n(n+1)} &= \frac{4n+4}{n(n+1)} \\ &= \frac{4(n+1)}{n(n+1)} \end{aligned}$$

We divide the numerator and the denominator by $(n+1)$. Since $n \geq 2$, $n+1 \neq 0$.

$$\frac{4(n+1)}{n(n+1)} = \frac{4}{n}$$

Thus, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{4}{n}$. This completes the proof. $\square$

Proof of Lemma 3.2. Let $n \in \mathbb{N}_{>1}$ be such that $n \equiv 3 \pmod{4}$. By definition, there exists an integer $k \geq 0$ such that $n = 4k + 3$. We define $x = \frac{n+1}{4}$. Substitute $n = 4k + 3$:

$$\begin{aligned} x &= \frac{(4k+3)+1}{4} \\ &= \frac{4k+4}{4} \\ &= k+1 \end{aligned}$$

Since $k \geq 0$, $x \in \mathbb{N}_{>0}$. We define $y = \frac{n(n+1)}{2}$. Substitute $n = 4k + 3$:

$$\begin{aligned} y &= \frac{(4k+3)(4k+4)}{2} \\ &= \frac{(4k+3)4(k+1)}{2} \\ &= 2(4k+3)(k+1) \end{aligned}$$

Since $k \geq 0$, $4k + 3 \geq 3$ and $k + 1 \geq 1$. Both are integers, so their product multiplied by 2 is an integer, and $y \in \mathbb{N}_{>0}$. We define $z = \frac{n(n+1)}{2}$. Since this expression is identical to y , $z \in \mathbb{N}_{>0}$.

We now compute the sum $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$:

$$\begin{aligned}\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{\frac{n+1}{4}} + \frac{1}{\frac{n(n+1)}{2}} + \frac{1}{\frac{n(n+1)}{2}} \\ &= \frac{4}{n+1} + \frac{2}{n(n+1)} + \frac{2}{n(n+1)}\end{aligned}$$

Summing the last two fractions:

$$\frac{2}{n(n+1)} + \frac{2}{n(n+1)} = \frac{4}{n(n+1)}$$

Now, sum with the first fraction:

$$\frac{4}{n+1} + \frac{4}{n(n+1)}$$

Find the common denominator $n(n+1)$:

$$\frac{4}{n+1} = \frac{4n}{n(n+1)}$$

Summing the numerators:

$$\frac{4n+4}{n(n+1)} = \frac{4(n+1)}{n(n+1)}$$

Divide the numerator and denominator by $(n+1)$. Since $n \geq 2$, $n+1 \neq 0$:

$$\frac{4(n+1)}{n(n+1)} = \frac{4}{n}$$

Thus, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{4}{n}$. This completes the proof. □

5 Architecture for Autoformalization in Lean 4

To facilitate automated verification by a formal system such as Lean 4, the proofs can be codified as follows.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Nat.Parity

/--
The Erdos-Straus Conjecture
-/
def erdos_straus_conjecture : Prop := ∀
  n : ℕ, n > 1 → ∃ x y z : ℕ, x > 0 ∧ y > 0 ∧ z > 0 ∧
  4 * x * y * z = n * (x * y + x * z + y * z)

/--
Lemma 1: Solution for n = 2 mod 3
-/
theorem erdos_straus_mod3_2 (n : ℕ) (h1 : n > 1) (h2 : n % 3 = 2) : ∃
  x y z : ℕ, x > 0 ∧ y > 0 ∧ z > 0 ∧
  4 * x * y * z = n * (x * y + x * z + y * z) := by
```

```
    admit

/--
Lemma 2: Solution for  $n = 3 \bmod 4$ 
-/
theorem erdos_straus_mod4_3 (n :  $\mathbb{N}$ ) (h1 :  $n > 1$ ) (h2 :  $n \% 4 = 3$ ) :  $\exists$ 
  x y z :  $\mathbb{N}$ ,  $x > 0 \wedge y > 0 \wedge z > 0 \wedge$ 
     $4 * x * y * z = n * (x * y + x * z + y * z)$  := by
  admit
```